CENTER FOR COMPETITIVE STATECRAFT AND STRATEGIC POLICY

WP-2026 SERIES

SYSTEM RESILIENCE AND IMPACT ANNEX

30/90/180-Day Cascade Modeling — UECC, ChMP, Isotope JSC Disruption Scenarios

NOTICE *Independent policy research. All sources open-source and cited at point of use. This document represents independent analytical assessment, does not constitute official policy direction or legal advice, and does not substitute for interagency consequence analysis.*

KEY JUDGMENTS

KJ 1

Russia's three highest-leverage enforcement targets — UECC (enrichment), ChMP (conversion/zirconium), and Isotope JSC (radioisotopes) — face materially different system resilience constraints. UECC disruption would trigger a global enrichment supply shock within 90 days that Western alternative capacity cannot currently absorb. ChMP disruption can be sequenced to allow alternative supply development. Isotope JSC disruption carries negligible reactor safety risk and is immediately executable without system consequence.

ALMOST CERTAINLY — HIGH CONFIDENCE

KJ 2

Western alternative enrichment capacity (Urenco, Orano, Centrus) is sufficient to replace Russian supply for VVER-1000 and Western-design reactors within 3–5 years under current expansion trajectories, but is insufficient to absorb a sudden UECC disruption within the 12–18 month fuel cycle replacement window. The supply precondition for UECC enforcement action (allied capacity ≥85% of UECC displaced volume) is not yet met as of March 2026.

ALMOST CERTAINLY — HIGH CONFIDENCE — BASED ON T1 SOURCES

KJ 3

VVER-440 reactor fuel is now available from Westinghouse/Enusa for Finland (Loviisa, first reload September 2024), Czech Republic (Dukovany, deliveries from 2024), and Slovakia. Hungary (Paks) remains the principal holdout due to Paks II construction dependency on Rosatom. The VVER-440 fuel supply diversification timeline is 2024–2027; the VVER-440 supply precondition for enforcement action against ChMP is partially met.

ALMOST CERTAINLY — HIGH CONFIDENCE — T1 SOURCES

I. ANALYTICAL FRAMEWORK — CASCADE MODELING METHODOLOGY

This annex models the 30-, 90-, and 180-day consequences of enforcement action against the three highest-scoring Tier 1 nodes in the SENI scoring model. For each node, the analysis identifies: (a) Russian fuel cycle redundancy — can Russia compensate from other nodes; (b) Western dependency exposure — which allied reactor operators face supply risk; and (c) alternative supply timeline — when can replacement supply come online. This analysis anchors D3 (Collateral Impact) and D5 (Time-to-Impact) scores to modeled timelines rather than categorical estimates.

II. SCENARIO A — UECC (N28) DISRUPTION: ENRICHMENT SUPPLY CASCADE

ATTRIBUTION	UECC (Novouralsk) accounts for 48% of Russian enrichment capacity and approximately 20% of global SWU capacity [TIER 2 — World Nuclear Association]. Russia's total enrichment capacity is approximately 27 million SWU/year; UECC's share is approximately 11–13 million SWU/year. Global demand is approximately 47–50 million SWU/year [TIER 2 — Mordor Intelligence 2024; TIER 2 — ScienceDirect 2025]. Urenco and Orano together account for 41% of total global capacity [TIER 2 — CEE Energy News, September 2025]. EU dependence on Russian enrichment was 23% in 2024, down from 38% in 2023 [TIER 2 — CEE Energy News].
JUDGMENT	Available reporting supports a high-confidence judgment that sudden UECC disruption would create a global enrichment supply deficit of approximately 11–13 million SWU/year that Western alternative capacity cannot absorb within a single fuel cycle. The 90-day window is critical: utilities typically maintain 12–18 months of SWU forward coverage, meaning the operational impact of a sudden disruption would be felt in reactor refueling schedules beginning 12–18 months after enforcement action.
LIMITS	The specific contract positions of individual Western utilities with respect to Russian SWU supply are not publicly disclosed. Some utilities may have 2–3 year forward coverage providing a longer buffer. The model uses industry-average assumptions rather than utility-specific positions.

DIMENSION	DAY 0–30	DAY 31–90	DAY 91–180	LONG-TERM (>180 DAYS)
Russian redundancy	Zero — UECC cannot be compensated from other Russian sites without	Zero — Russian enrichment capacity cannot be rapidly redistributed to other	Partial — Zelenogorsk (ECP) capacity available but insufficient to replace UECC	Partial — Russia can compensate approximately 30–40% of UECC volume from other sites within 2–3 years

	multi-year centrifuge installation	sites	volume	
Global SWU deficit	~11–13M SWU/yr suddenly unavailable	Utilities begin drawing down forward contracts; spot market price surge	Forward coverage exhausted for utilities without 18+ month buffer; first fuel delivery failures	Supply deficit persists until Western capacity expansion completes
Western utility impact	Utilities with <12 months forward coverage face immediate contract gap	Utilities begin emergency procurement; spot price surge 50–200%	First reactor outages possible for utilities without adequate cover; Eastern European VVER operators most exposed	Structural supply gap until 2028–2030 Western capacity expansion completes. NOTE: This model uses industry-average assumptions. Utility-specific contract exposure (forward coverage duration, hedging positions) is not publicly disclosed and is not modeled. Macro accuracy is maintained; micro allocation risk requires classified energy contract data.
Price signal	Spot SWU price +20–50%	Spot SWU price +100–200%; long-term contract price pressure	Spot SWU price +200–400%; government intervention likely	Price normalization as new capacity comes online 2028–2032
Allied response	Emergency DOE/NNSA review; five eyes nuclear contingency coordinati	EU emergency fuel security consultation; IAEA safeguards implication	Congressional and European Parliament political response; potential UECC enforceme	Long-term: Western enrichment expansion accelerated; Russian market share permanently reduced

	on	s assessed	nt reversal pressure	

A. Precondition Analysis — Why UECC is Tier 2 Not Tier 1

The SENI scoring model places UECC in Tier 2 due to the D3=1 hard-stop (catastrophic collateral) and the PRECONDITION UNMET dominant constraint. This cascade model validates both. The precondition — allied enrichment capacity ≥85% of UECC displaced volume — requires approximately 9.4–11 million additional Western SWU/year before UECC disruption is manageable. Current Western expansion programs (Urenco Netherlands/UK/Germany, Orano Tricastin, Centrus Piketon) are projected to add 5–8 million SWU/year by 2030 [TIER 2 — market analysis sources]. The precondition is partially met as of 2026; it becomes more fully met by 2028–2030.

DOCUMENTED STRUCTURAL CONSTRAINT ENFORCEMENT SEQUENCING IMPLICATION: UECC enforcement action is premature before 2028 under current Western capacity expansion trajectories unless an emergency allied fuel security program accelerates the timeline. The system resilience analysis confirms the scoring model's Tier 2 classification and the PRECONDITION UNMET dominant constraint.

B. Russian Redundancy — Enrichment System Resilience

RUSSIAN ENRICHMENT NODE	CAPACITY (approx)	UECC COMPENSATION CAPABILITY
UECC (Novouralsk) — DISRUPTED	~11–13M SWU/yr	N/A — disruption scenario
ECP (Zelenogorsk)	~8.7M SWU/yr	Partial compensation; already near full utilization for foreign orders
SCC (Seversk)	~Moderate capacity	Primarily handles reprocessed uranium; not configured for commercial foreign orders
AECC (Angarsk)	~3M SWU/yr	IUEC multilateral structure; cannot be rapidly repurposed
Total Russian without UECC	~15–17M SWU/yr	Insufficient for current foreign order book without UECC (~20M SWU/yr required)

Russia cannot fully compensate for UECC disruption from its remaining enrichment capacity. Foreign order book requirements without UECC would exceed remaining Russian capacity, forcing Russia to choose between domestic reactor supply and foreign contract fulfillment. This compounds the enforcement effect — Russia faces its own supply crisis if UECC is disrupted.

III. SCENARIO B — CHMP (N34) DISRUPTION: CONVERSION AND ZIRCONIUM CASCADE

ATTRIBUTION	ChMP (Glazov) is Russia's only zirconium alloy manufacturer and is responsible for fuel that operates in approximately one in six global reactors [TIER 1 — ChMP website self-disclosure]. ChMP is also a UF6 conversion node. TVEL was designated by OFAC in April 2024; ChMP as TVEL subsidiary is constructively covered but not explicitly listed. Westinghouse/Enusa supply VVER-440 fuel with deliveries beginning 2024 to Finland (Loviisa) and Czech Republic (Dukovany) [TIER 1 — Westinghouse press releases 2024].
JUDGMENT	Available reporting supports a high-confidence judgment that ChMP disruption can be sequenced to minimize reactor safety risk because (a) zirconium alloy cladding has a 12–18 month manufacturing lead time, giving advance warning, and (b) alternative zirconium suppliers exist (Sandvik, ATI in the U.S., Cezus/Framatome in France) but would require qualification and capacity expansion. The key variable is the VVER-440 fuel supply transition: operators with Westinghouse contracts (Finland, Czech Republic, Slovakia) have reduced their ChMP dependency; Hungary (Paks) remains the principal holdout.
LIMITS	The specific inventory positions of ChMP zirconium alloy held by Western fuel assembly manufacturers and the lead times for alternative qualification are not publicly documented with precision. The 12–18 month manufacturing lead time estimate is industry-standard but utility-specific positions vary.

DIMENSION	DAY 0–30	DAY 31–90	DAY 91–180	LONG-TERM (>180 DAYS)
Russian redundancy	Zero — ChMP is Russia's only zirconium producer	Zero — no Russian alternative exists	Zero for new production; existing ChMP inventory partially compensates	Russia faces domestic fuel assembly production risk without zirconium supply

VVER-440 fuel	No immediate impact — Westinghouse/Enusa deliveries ongoing for Finland/Czech Republic/ Slovakia	Fuel assembly manufacturers assess zirconium inventory; TVEL begins rationing	TVEL cannot complete Western customer fuel orders without zirconium; contract defaults begin	Western VVER-440 operators (Finland, Czech Republic, Slovakia) insulated; Hungary/Paks exposed
Western-design reactor fuel	Sandvik (Sweden), ATI (USA), Cezus (France) are alternative zirconium sources; 12–18 month qualification required	Alternative qualification accelerated; emergency procurement begins	Alternative sources partially operational; some delivery delays	Alternative zirconium supply modeled as fully operational within 18–24 months. NOTE: This assumes successful qualification scaling without regulatory delay, manufacturing bottleneck, or geopolitical disruption. The model captures availability of alternative sources, not execution risk of the substitution process. Qualification failure or delay would extend the cascade timeline.
Conversion impact	ChMP dual function: UF6 conversion partially disrupted	Other conversion nodes (SCC, AECC) compensate partially	Conversion disruption cascades to enrichment in 3–6 months	Full conversion disruption impact felt in reactor fuel supply 6–12 months post-disruption

A. VVER-440 Transition Status — Which Operators are Insulated

OPERATOR / COUNTRY	VVER-440 STATUS	ChMP RISK EXPOSURE
Fortum / Loviisa, Finland	Westinghouse VVER-440 first reload September 2024; long-term partnership [T1 —	LOW — Westinghouse supply operational; Russian dependency ended

	Westinghouse PR 2024]	
ČEZ / Dukovany, Czech Republic	Westinghouse 7-year contract; deliveries from 2024 [T1 — Westinghouse PR 2023]	LOW — Westinghouse supply operational; transition underway
Slovenské Elektrárne / Slovakia	ESSANUF project participation; transition underway	MODERATE — transition in progress; some Russian dependency remains
Paks / Hungary	Paks II construction contract with Rosatom; structural dependency on TVEL fuel supply	HIGH — Hungary remains dependent on TVEL/ChMP; Paks II construction creates long-term lock-in
MVM / Hungary (Paks I)	Paks I fleet continues Russian fuel; no announced Westinghouse transition	HIGH — Paks I still Russian fuel dependent

DOCUMENTED STRUCTURAL CONSTRAINT Hungary is the critical residual exposure. EU enforcement action against ChMP would trigger a fuel supply crisis for Hungary's Paks nuclear fleet, which provides approximately 50% of Hungary's electricity. This is the primary D3 constraint for ChMP enforcement action in the EU context. U.S. OFAC enforcement action against ChMP is less constrained by Hungarian reactor dependency but secondary sanctions on European banks clearing ChMP payments could affect Hungarian operators indirectly.

IV. SCENARIO C — ISOTOPE JSC (N33/N43) DISRUPTION: NO SYSTEM CASCADE

ATTRIBUTION	Isotope JSC distributes radioisotopes manufactured at NIIAR, ECP, NCCP, and other Rosatom facilities [TIER 1 — Isotope JSC website]. Its primary Western-relevant products are Mo-99 (medical imaging), Cf-252 (oil well logging), and Li-6 (tritium production). None of these products are reactor fuel; disruption of Isotope JSC does not affect reactor operations. Alternative Mo-99 suppliers include SHINE Medical (Wisconsin, FDA approval pending as of 2024), Laurentis/BWXT (Canada, Darlington production beginning 2024), and European producers (IRE Belgium, NTP South Africa) [TIER 2 — SHINE; Laurentis; industry sources].
JUDGMENT	Available reporting supports a high-confidence judgment that Isotope JSC disruption carries no reactor safety risk and minimal systemic consequence. Medical isotope supply disruption from Isotope JSC would be partially compensated by alternative

	Western and Australian producers within 30–90 days, with SHINE and Laurentis/BWXT domestic North American capacity providing near-term buffer. This confirms D3=5 (no reactor safety collateral) and validates the Tier 1 immediate action classification.
LIMITS	The medical isotope supply chain is fragile due to short half-lives (Mo-99 half-life 66 hours). Disruption events have historically caused clinical scanning shortages. However, the fragility is a logistics and production coordination problem, not a ChMP/UECC-style structural dependency problem. Alternative producers can compensate within weeks to months rather than years.

DIMENSION	DAY 0–30	DAY 31–90	DAY 91–180	LONG-TERM (>180 DAYS)
Medical isotopes (Mo-99/Tc-99m)	Short-term shortage possible; existing inventory buffers initial disruption	SHINE, Laurentis/BWXT, IRE Belgium increase production to compensate	North American domestic supply operational; shortage resolved	Russian Mo-99 market share permanently reduced; domestic alternatives established
Reactor safety impact	NONE — Mo-99 is not reactor fuel	NONE	NONE	NONE — Isotope JSC is not in the reactor fuel supply chain
Cf-252 (oil well logging)	ISOFLEX USA and other distributors assess inventory; alternative global sources (ORNL, neutron generator technology)	U.S. Cf-252 production from Oak Ridge (ORNL) can partially compensate; neutron generators substitute for some applications	Alternative sources operational; Cf-252 shortage manageable	Cf-252 gap filled by ORNL production and neutron generator alternatives

Li-6 / tritium (N33 NCCP)	No immediate reactor impact; Li-6 is weapons/tritium production material, not civilian reactor fuel	No civilian reactor impact	No civilian reactor impact	DPRK proliferation nexus is disrupted — this is the enforcement objective, not reactor supply

DOCUMENTED STRUCTURAL CONSTRAINT KEY FINDING: Isotope JSC disruption has no reactor safety consequence and minimal civilian supply consequence (medical isotopes can be sourced from Western alternatives). This validates D3=5 (no collateral) and confirms Tier 1 immediate action classification. The scoring model's identification of N33 and N43 as the highest-priority enforcement actions in the index is vindicated by the cascade analysis — they are the nodes where enforcement effect is highest and system disruption risk is lowest.

V. CROSS-SCENARIO SUMMARY — ENFORCEMENT SEQUENCING GUIDANCE

NODE	TIER	D3 CASCADE RISK	IMMEDIATE ACTION?	SEQUENCING GUIDANCE
N43/N33 Isotope JSC	TIER 1	NONE — no reactor safety consequence	YES — immediate	Execute first. No precondition. No system risk. Designation package predicate established (FPA-01 v3).
N34 ChMP	TIER 1	MODERATE — Hungary/Paks residual; 12–18 month mitigation window	YES — with monitoring	Execute after Hungary diplomatic engagement. GL 115A expiry (June 30 2025) is the trigger. Westinghouse VVER-440 transition to Finland/Czech Republic/Slovakia provides cover.
N28 UECC	TIER 2	CATASTROPHIC — global enrichment	NO — precondition unmet	Defer until Western enrichment capacity ≥85% of UECC displaced volume. Estimated 2028–2030. Parallel preparation: designation package ready;

		supply shock		allied consultation complete.
N30 ECP / Zelenogorsk	TIER 2	MODERATE — enrichment + Isotope JSC gap	CONDITIONAL	Pair with Isotope JSC designation. ECP isotope operations independent of enrichment supply consequences.
N47 Bochvar VNIINM	TIER 2	LOW — MOX fabrication; no operating reactor fuel supply consequence	CONDITIONAL	Designation signals Pu pathway enforcement without reactor supply risk. Pair with Mayak/Zheleznogorsk RT-2 procurement targeting.

VI. MODELING LIMITATIONS — IRREDUCIBLE OSINT CONSTRAINTS

The following limitations are inherent to open-source cascade modeling and cannot be resolved without classified or restricted-access data. They are disclosed here for interagency review purposes.

LIMITATION	ANALYTICAL IMPACT
Macro vs. micro utility exposure (UECC)	The UECC cascade model uses industry-average assumptions for forward SWU coverage (12–18 months) and spot price response. Actual impact depends on utility-specific contract positions that are not publicly disclosed. The model correctly identifies system-level disruption magnitude but cannot allocate risk across individual reactor operators. Utility-specific impact requires classified energy contract data.
Zirconium substitution execution risk (ChMP)	The ChMP cascade assumes alternative suppliers (Sandvik, ATI, Cezus) can qualify and scale production within 18–24 months. This models availability of alternatives, not execution risk. Qualification failure, regulatory delay, manufacturing bottleneck, or supply chain disruption could extend the cascade timeline materially. These risks are not modeled probabilistically — they are acknowledged as scenario assumptions.
Russian response modeling	Russian fuel cycle redundancy is modeled from open-source capacity data. Russian government decisions about domestic vs. export allocation in a disruption scenario are not modeled — these depend on political and strategic factors not accessible through open-source analysis.
Spot market price response	Price surge estimates (+50–400%) are directional based on supply-demand gap analysis, not derived from energy market econometric models. Actual price response depends on inventory drawdown, contract vs. spot market

	balance, and government intervention dynamics.

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CHINA FINANCIAL ESCALATION LADDER

Tiered Model for Russia-China Nuclear Fuel Cycle Financial Engagement — Annex to WP-2026-SENI-FIN-01

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KEY JUDGMENTS

KJ 1

Post-2022 CNY migration has substantially reduced U.S. financial leverage over the Russia-China nuclear fuel relationship. ICBC and Bank of China are confirmed as the primary Chinese state bank channels for Russia-China nuclear transactions. Secondary sanctions on either institution would constitute a G-SIB-level financial escalation with systemic consequences extending well beyond the nuclear fuel sector.

ALMOST CERTAINLY — HIGH CONFIDENCE — T1 SOURCES

KJ 2

China's role in the global uranium enrichment market is strategically ambiguous: Chinese state firms are increasing imports of Russian enriched uranium while simultaneously building domestic enrichment capacity. Flow data for 2024–2025 indicates displacement exchanges whereby China imports Russian enriched uranium for domestic use while exporting its own enriched uranium to Western markets — a mechanism that partially circumvents Western sanctions at the commodity level.

LIKELY — MODERATE-HIGH CONFIDENCE — T2 SOURCES

KJ 3

A tiered escalation ladder is required because the full range from diplomatic engagement to G-SIB designation represents qualitatively different policy instruments with radically different cost-benefit profiles. Skipping tiers is strategically counterproductive — diplomatic engagement and targeted entity action should be exhausted before financial institution action is considered.

ALMOST CERTAINLY — HIGH CONFIDENCE — ANALYTICAL JUDGMENT

I. THE POST-2022 CNY SETTLEMENT ARCHITECTURE

ATTRIBUTION	Congressional Research Service analysis documents the migration of Russia-China energy transactions to CNY settlement post-2022 [TIER 1 — CRS R47742]. Rosatom-CNNC nuclear fuel contracts have migrated to CNY settlement through Chinese state banks. ICBC and Bank of China are confirmed as primary clearing institutions for Chinese state enterprise energy transactions with Russia [TIER 2 — multiple financial analysis sources]. Rosatom's 2024 data shows China increasing to represent a substantially larger share of Russian enriched uranium exports as Western demand contracted [TIER 2 — industry reporting; TIER 2 — CEE Energy News]. Global uranium enrichment pricing reached $97.66/SWU average in 2024 [TIER 2 — market analysis].
JUDGMENT	Available reporting supports a high-confidence judgment that the Russia-China nuclear fuel relationship has become substantially financially insulated from U.S. enforcement instruments post-2022. CNY settlement removes the USD correspondent banking nexus that is the primary mechanism for OFAC enforcement. Secondary sanctions leverage exists but requires designating Chinese financial institutions, which creates G-SIB-level systemic risk.
LIMITS	The specific contract terms, volumes, and clearing mechanisms for CNNC-TVEL nuclear fuel transactions are not publicly disclosed. The displacement exchange hypothesis (China imports Russian uranium for domestic use; exports Chinese uranium to Western markets) is based on trade flow data and analysis, not confirmed documentation of the mechanism.

II. ESCALATION LADDER — SEVEN TIERS

The following tiered model provides a structured framework for engaging the China-Russia nuclear fuel financial relationship from the least to the most disruptive intervention. Each tier is defined by its mechanism, cost-benefit profile, trigger conditions, and prerequisites.

TIER 1: Diplomatic Engagement — Bilateral Nuclear Regulatory

Mechanism	State Department diplomatic channel; IAEA Board of Governors engagement; bilateral U.S.-China nuclear energy dialogue
Cost-Benefit	Low cost, uncertain benefit. Establishes formal record of U.S. concerns before more coercive measures. Required for any subsequent tier to have political legitimacy.
Trigger	Active upon any confirmed CNNC-TVEL commercial transaction for controlled materials.
Prerequisite	None — available immediately.

Effect	Signals concern; creates diplomatic record; no financial disruption.

TIER 2: BIS Entity List — Targeted Technology Controls

Mechanism	BIS Entity List additions for specific CNNC subsidiaries or TVEL-CNNC joint venture entities involved in controlled material transactions. ECCN licensing requirements applied to controlled equipment and technology.
Cost-Benefit	Moderate cost (Chinese retaliation possible on tech exports), targeted benefit. Restricts specific technology transfer pathways without broader financial disruption.
Trigger	Confirmed CNNC participation in TVEL-sanctioned activities or acquisition of U.S.-origin controlled technology for nuclear programs inconsistent with NPT obligations.
Prerequisite	BIS review and interagency coordination; State Department concurrence.
Effect	Technology access restriction; modest deterrence signal; no systemic financial impact.

TIER 3: Targeted Secondary Sanctions — CNNC Subsidiaries and SPVs

Mechanism	OFAC designation under EO 13382 or EO 14024 (Russia nexus) of specific CNNC subsidiaries or joint venture SPVs that directly transact with designated Russian entities (TVEL). Targets the specific commercial entities, not Chinese state banks.
Cost-Benefit	Moderate-high cost (bilateral tension, potential Chinese countermeasures on SOE transactions), high targeted benefit. Disrupts specific Russia-China nuclear commercial channels without G-SIB exposure.
Trigger	Documented CNNC subsidiary material support to TVEL post-TVEL SDN designation (April 2024). EO 14024 Section 11 determination that CNNC operations in Russian nuclear fuel sector constitute Russian military-industrial base support.
Prerequisite	Interagency clearance; NSC review; allied notification; trade intelligence confirming designated-entity nexus.
Effect	Disrupts specific Russia-China nuclear commercial relationship; signals to other Chinese entities the secondary sanctions risk; does not affect Chinese state

	banking operations broadly.

TIER 4: CAATSA §231 Defense Sector Determination — Nuclear Programs

Mechanism	State Department CAATSA §231 determination that CNNC has engaged in significant transactions with Russian defense sector entities, triggering menu-based sanctions options up to and including asset freezes.
Cost-Benefit	High cost (bilateral defense and economic relationship implications), high benefit if executed with allied coordination. Requires Congressional notification.
Trigger	Documented CNNC significant transaction with Russian defense sector; established nexus to proliferation-relevant Russian nuclear programs. CAATSA §231 standard: 'significant transaction with a sanctioned person.'
Prerequisite	Congressional notification (30 days); NSC/principals-level decision; potential waiver discussion for critical national security interests.
Effect	Asset freeze exposure for designated CNNC entities; chilling effect on Chinese state enterprise engagement with designated Russian entities broadly.

TIER 5: Targeted Chinese Bank Restrictions — Non-Systemic

Mechanism	OFAC CAPTA Directive or secondary sanctions on specific Chinese bank subsidiaries or overseas branches (not parent institutions) that clear Russia-China nuclear transactions. Targeting bank branches in third-country jurisdictions (Hong Kong, Singapore) rather than mainland Chinese parent banks.
Cost-Benefit	High cost (escalation potential with Chinese banking sector), potentially high benefit for specific transaction channels. Avoids full G-SIB designation while restricting specific clearing channels.
Trigger	Identified specific branch or subsidiary of ICBC/Bank of China/China Development Bank demonstrably clearing Russia-TVEL nuclear transactions post-SDN designation.
Prerequisite	FININT confirming specific branch clearing nexus; Treasury GC legal review; NSC principals decision; allied consultation (impact on European USD clearing through Hong Kong branches).
Effect	Restricts specific clearing mechanism; signals escalation pathway toward parent

	bank; creates chilling effect on Chinese banking engagement with TVEL-adjacent transactions.

TIER 6: CAPTA Directive — Correspondent Account Restriction on Chinese G-SIBs

Mechanism	OFAC Correspondent Account or Payable-Through Account (CAPTA) Directive restricting U.S. correspondent account access for designated Chinese banks for specific categories of transactions. This is short of full blocking sanctions.
Cost-Benefit	Very high cost (potential for USD clearing disruption to global trade flows; Chinese retaliation against U.S. financial institutions in China), potentially transformative benefit. This action is structurally equivalent to the SWIFT restrictions applied to major Russian banks in 2022 — a categorically different magnitude of intervention.
Trigger	Demonstrated pattern of significant Chinese bank clearing of TVEL-designated transactions at scale; evidence of deliberate evasion of U.S. sanctions; NSC principals-level determination that benefits outweigh systemic risk.
Prerequisite	G7 allied coordination required; Treasury/Fed consultation on systemic financial stability implications; NSC principals decision; Presidential authorization. Timeline: 6–18 months minimum preparation.
Effect	Restricts USD correspondent account access for targeted Chinese banks; potentially disrupts Russian-Chinese nuclear fuel trade substantially; creates significant risk of broader financial market disruption.

TIER 7: Full SDN Designation — Chinese G-SIB

Mechanism	Full OFAC SDN blocking designation of ICBC, Bank of China, or China Development Bank under EO 14024.
Cost-Benefit	Potentially catastrophic cost (equivalent to major global financial crisis trigger; Chinese retaliation against U.S. treasury holdings and financial infrastructure; potential de-dollarization acceleration), maximum theoretical benefit. This tier is essentially equivalent to declaring financial war on China.
Trigger	This tier has no analytically defensible trigger in the nuclear enforcement context. The benefits of disrupting Russian nuclear fuel transactions would not justify the systemic cost under any foreseeable scenario.
Prerequisite	Not analytically applicable. This tier is included for completeness and to

	demonstrate why it is off the table.
Effect	ANALYTICALLY OFF THE TABLE. Systemic global financial disruption would dwarf any nuclear enforcement benefit.

III. ANALYTICAL ASSESSMENT — PRACTICAL ESCALATION RANGE

The analytically defensible escalation range for the nuclear fuel enforcement context is Tiers 1–5. Tier 6 (CAPTA Directive) is a theoretical option that would require extraordinary circumstances and full G7 coordination. Tier 7 (Full G-SIB SDN) is off the table under any analytically defensible nuclear enforcement rationale.

The practical effective range is Tiers 1–4. Tier 3 (targeted CNNC subsidiary secondary sanctions) is the highest-leverage action that avoids systemic financial risk. The transition from Tier 3 to Tier 4 (CAATSA §231 determination) represents a qualitative escalation that requires Congressional notification and should be reserved for demonstrated significant violations.

IMPLEMENTATION FRICTION The hostile review critique '#9 — escalation risk under-modeled (China)' identified that recommending secondary sanctions on Chinese banks without an escalation ladder was strategically naive. This tiered model directly addresses that critique by demonstrating that (a) Tier 3 targeted entity action is available without G-SIB exposure; (b) Tiers 6–7 are correctly categorized as disproportionate responses; and (c) the escalation pathway from engagement to coercive instruments is sequential, not simultaneous.

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SANCTIONS AUTHORITY ANNEX

Evidentiary Thresholds, Legal Standards, and Allied Divergence — WP-2026-SENI Enforcement Authorities

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KEY JUDGMENTS

KJ 1

The existence of legal authority under EO 13382, EO 14024, or CAATSA is a necessary but not sufficient condition for enforcement action. Designation packages also require: a traceable financial nexus, evidentiary documentation meeting OFAC standards, and a factual basis for the designation criterion. The SENI corpus meets these standards for N43/N33 (Isotope JSC pathway); additional financial intelligence development is required for most Tier 2 nodes.

ALMOST CERTAINLY — HIGH CONFIDENCE — ANALYTICAL JUDGMENT

KJ 2

Allied legal divergence — specifically the EU blocking statute (Regulation 2271/96) and the EU-Russia nuclear fuel carve-outs — means that U.S. secondary sanctions targeting Russian nuclear fuel entities will produce different legal obligations for EU operators than for U.S. persons. This divergence must be managed through allied coordination before secondary sanctions are activated at scale.

ALMOST CERTAINLY — HIGH CONFIDENCE — T1 SOURCES

I. PRIMARY ENFORCEMENT AUTHORITIES — THRESHOLD ANALYSIS

AUTHORITY	DESIGNATION CRITERION	EVIDENTIARY THRESHOLD FOR SENI NODES
EO 13382 — WMD Proliferators	(i) Engaged in activities to transfer WMD or delivery systems; (ii) provided support to	Isotope JSC: met — Rosatom subsidiary chain; Cf-252 (1C236) and Li-6 (1C233) WMD-relevant materials. ChMP: met — zirconium cladding in nuclear fuel supply chain; TVEL SDN constructive

	such entities; (iii) owned/controlled by or acting for such entities	coverage. Named financial nexus (Nordion/Lantheus/Wells Fargo) confirms U.S. person nexus for designation package.
EO 14024 — Russia Harmful Activities	Operating or having operated in the defense and related materiel or technology sector of the Russian Federation economy	TVEL April 2024 designation covers TVEL entity. TVEL subsidiary designation (ChMP, NCCP) derivable through 50% ownership rule. Independent designation of subsidiaries not explicitly listed requires specific determination.
CAATSA §231	Significant transaction with a sanctioned person in defense sector	Russian nuclear fuel sector: Rosatom/TVEL designated persons. CAATSA §231 applicable to non-U.S. persons engaging in significant transactions with designated TVEL entities. Significant transaction standard requires volume and context analysis.
EO 13722 — DPRK Sanctions	Material support to DPRK; transactions with DPRK government or WPK	N33 NCCP Li-6/DPRK nexus: LOW CONFIDENCE — ANALYTICALLY INFERRED. Not sufficient for EO 13722 designation without additional evidentiary development. N36 Vostochny/FESCO: UN Panel documentation supports EO 13722 nexus for DPRK corridor.
BIS EAR — Export Controls	ECCN 1C236 (Cf-252), 1C233 (Li-6), 810.9 (zirconium cladding)	BIS Entity List designation available for any entity operating without adequate end-use controls on controlled materials. Kurchatov Institute, Sarov VNIIEF: BIS Entity List based on WMD-related research. ChMP: ECCN 810.9 controls on nuclear-grade zirconium.

II. OFAC DESIGNATION PACKAGE — EVIDENTIARY STANDARDS

An OFAC SDN designation package under EO 13382 requires: (a) factual basis for the designation criterion; (b) nexus documentation — connection between the target entity and U.S. persons, U.S.-owned property, or U.S. correspondent banking; (c) financial documentation supporting the designation; and (d) Treasury General Counsel legal review confirming authority and evidentiary sufficiency. This analysis does not constitute legal advice on any specific designation package.

A. Isotope JSC — Designation Package Elements

ELEMENT	SENI CORPUS STATUS
Factual basis — WMD-relevant	Cf-252 (ECCN 1C236) and Li-6 (ECCN 1C233): documented at

materials	Tier 1 through Isotope JSC product pages and NCCP/NIIAR production documentation. Rosatom subsidiary chain: documented at Tier 1. STRENGTH: HIGH.
Nexus — U.S. person connection	ISOFLEX USA (San Francisco) as U.S. registered distributor of Russian isotopes: documented at Tier 2. Lantheus Medical Imaging (Billerica MA, NYSE: LNTH) as named U.S. Mo-99 customer through Nordion: documented at Tier 1. Wells Fargo Bank N.A. as named U.S. correspondent bank: documented at Tier 1 (SEC S-1). STRENGTH: HIGH.
Financial documentation	Nordion supply agreement (2010–2020): Tier 1 PR Newswire documentation. Lantheus supply contracts: Tier 1 press releases. Wells Fargo banking relationship: Tier 1 SEC filing. Transaction-level SWIFT records: not available in OSINT. STRENGTH: MODERATE-HIGH (sufficient for designation predicate; transaction-level confirmation recommended).
Designation criterion — nexus to WMD proliferation	Isotope JSC distributes Cf-252 (ECCN 1C236 — potential use in neutron initiators) and Li-6 (ECCN 1C233 — tritium production for thermonuclear weapons) through uncontrolled channels. This meets the support criterion under EO 13382 §1(a)(ii). STRENGTH: HIGH for Cf-252 and Li-6; additional documentation needed for Cf-252 end-use verification gap.

III. ALLIED LEGAL DIVERGENCE — EU BLOCKING STATUTE

ATTRIBUTION	EU Regulation 2271/96 (the EU Blocking Statute) prohibits EU operators from complying with specified foreign extraterritorial sanctions, including OFAC secondary sanctions targeting EU business relationships with non-U.S. entities [TIER 1 — EU Regulation 2271/96]. U.S. secondary sanctions targeting European banks clearing TVEL-related transactions post-GL 115A expiry could create a conflict: the U.S. would require European banks to cease clearing; EU law would prohibit compliance with that requirement. The EU Blocking Statute applies to U.S. secondary sanctions that target EU entities' third-country business.
JUDGMENT	Available documentation supports a high-confidence judgment that U.S. secondary sanctions targeting European banks clearing TVEL transactions would trigger Regulation 2271/96 blocking statute obligations for EU-incorporated banks, creating a direct regulatory conflict. This is the primary allied legal divergence risk for SENI enforcement actions and must be managed through diplomatic coordination before secondary sanctions are activated targeting European financial institutions.
LIMITS	The specific scope of Regulation 2271/96 application to the current Russia nuclear sanctions programs has not been fully adjudicated. EU Member State interpretation varies. The regulatory conflict may be manageable through bilateral diplomatic

channels if the U.S. and EU coordinate enforcement timing and scope.

DIVERGENCE DIMENSION	ANALYSIS
Primary sanctions (U.S. persons)	No divergence — U.S. persons must comply with OFAC designations regardless of EU blocking statute.
Secondary sanctions (non-U.S. persons)	EU Blocking Statute potentially applicable if U.S. secondary sanctions target EU banks for clearing third-country (Russia-China) transactions. EU banks face conflicting obligations: U.S. secondary sanctions require non-compliance with TVEL; EU law may require compliance with TVEL contracts.
GL 115A expiry	Post-June 30, 2025 expiry creates enforcement exposure for EU banks clearing post-expiry ChMP transactions. This is a U.S.-jurisdiction enforcement action (secondary sanctions on EU banks) that could trigger EU Blocking Statute obligations. Allied coordination before GL 115A expiry is the recommended mitigation.
UK OFSI divergence	UK Office of Financial Sanctions Implementation (OFSI) operates under UK sanctions regulations post-Brexit. UK designations parallel U.S. designations but are not identical. TVEL April 2024 OFAC designation should be confirmed in UK OFSI list. UK Blocking Statute (Extraterritorial Jurisdiction Act) is less developed than EU equivalent.

IV. MENS REA — CORPORATE ENTITY DESIGNATION

OFAC designation of corporate entities under EO 13382 does not require proof of specific intent (mens rea) in the criminal law sense — designation is a civil administrative action. However, OFAC documentation standards require a factual basis demonstrating that the entity engaged in, or materially supported, designated activities. For SENI targets:

NODE	MENS REA / KNOWLEDGE ANALYSIS
Isotope JSC	Knowledge of Cf-252's dual-use status is constructively presumed — Isotope JSC's product documentation explicitly references oil well logging applications while Cf-252 is simultaneously listed on ECCN 1C236. No separate subjective intent required for designation.
ChMP	Manufacture of nuclear-grade zirconium alloy cladding is by definition an activity contributing to nuclear fuel supply. Knowledge of application is constructive.

UECC / ECP / SCC	Enrichment services for civilian nuclear programs do not require WMD intent; the designation basis is the entity's role in the Russian nuclear fuel cycle and its support relationship to designated entities in the WMD-adjacent context.
CNNC subsidiaries (China, Tier 3)	For secondary sanctions under EO 14024, the 'significant transaction' standard requires establishing that the Chinese entity transacted with a designated Russian SDN, not that it had specific proliferation intent. Volume and pattern analysis is the evidentiary standard.

V. D1 SCORE CALIBRATION REVIEW — SCORING MODEL VALIDATION

This annex provides the legal basis for validating the D1 (Legal Predicate Strength) scores assigned in the SENI scoring model. The following confirms or adjusts the D1 scores based on this authority analysis.

NODE	PRIMARY AUTHORITY	D1 v2	D1 VALIDATED	NOTES
N43 NIIAR	EO 13382 / BIS 1C236	5	5	Specific authority (ECCN 1C236) uncontested. D1=5 confirmed.
N33 NCCP	EO 13382 / BIS 1C233	5	5	Independent Li-6 ECCN 1C233 predicate. D1=5 confirmed.
N34 ChMP	EO 13382 / TVEL SDN	4	4	TVEL 50% rule constructive coverage confirmed; explicit listing recommended. D1=4 confirmed.
N28 UECC	TVEL SDN / EO 13382	5	5	TVEL designation April 2024 explicit. D1=5 confirmed.
N47 Bochvar	EO 13382	4	4	Rosatom subsidiary chain and Pu research predicate. D1=4 confirmed.
N31 Angarsk	IUEC treaty constraint	3	3	IUEC structure constrains application. D1=3 confirmed.
N46 Manzhouli	No U.S. authority	0	0	Chinese sovereign territory. D1=0 confirmed.

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